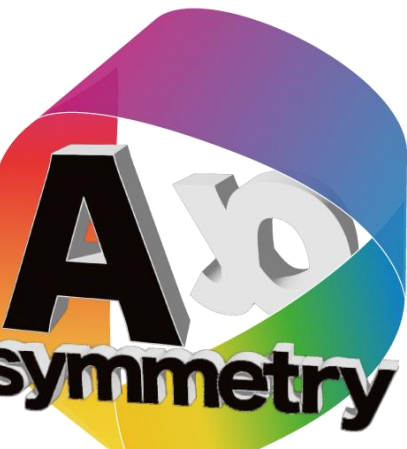




Uniaxial-Strain Effects on Antiferroquadrupolar Ordering and the Superconducting Transition in Cubic $\text{PrIr}_2\text{Zn}_{20}$



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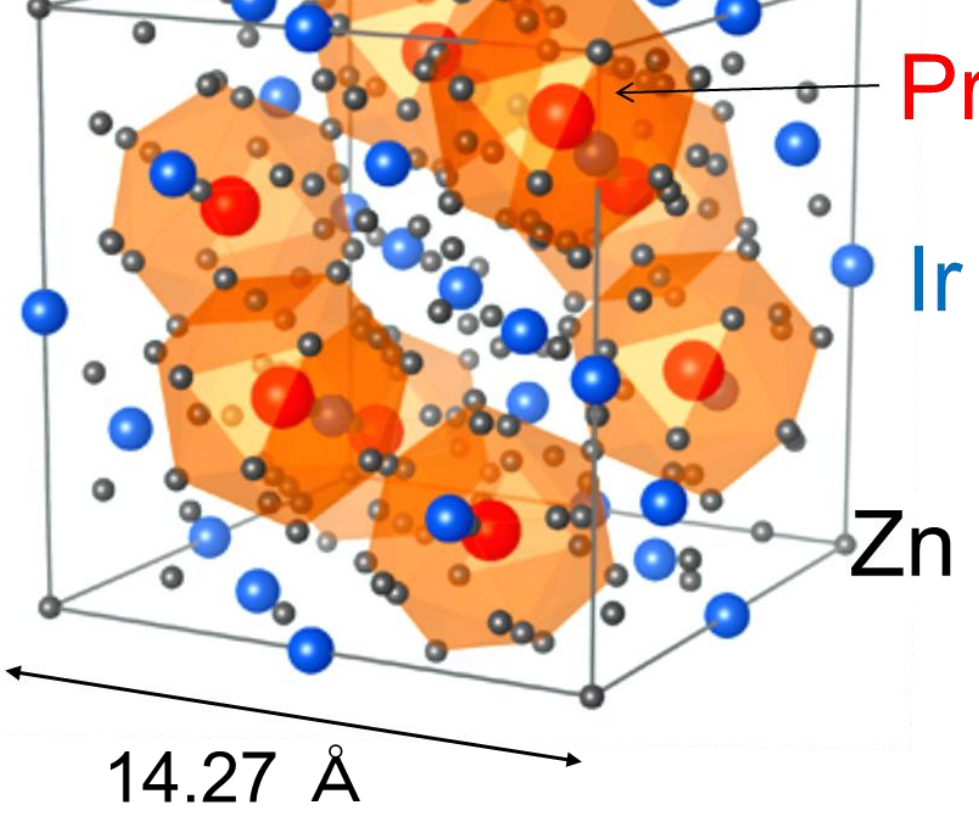
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Introduction

Crystal Structure & Ground State of $\text{PrIr}_2\text{Zn}_{20}$

Cubic $\text{CeCr}_2\text{Al}_{20}$ -Type [1]

Pr^{3+} ions in a **non-Kramers Γ_3 doublet ground state**, possess **electric quadrupolar degrees of freedom** [2].

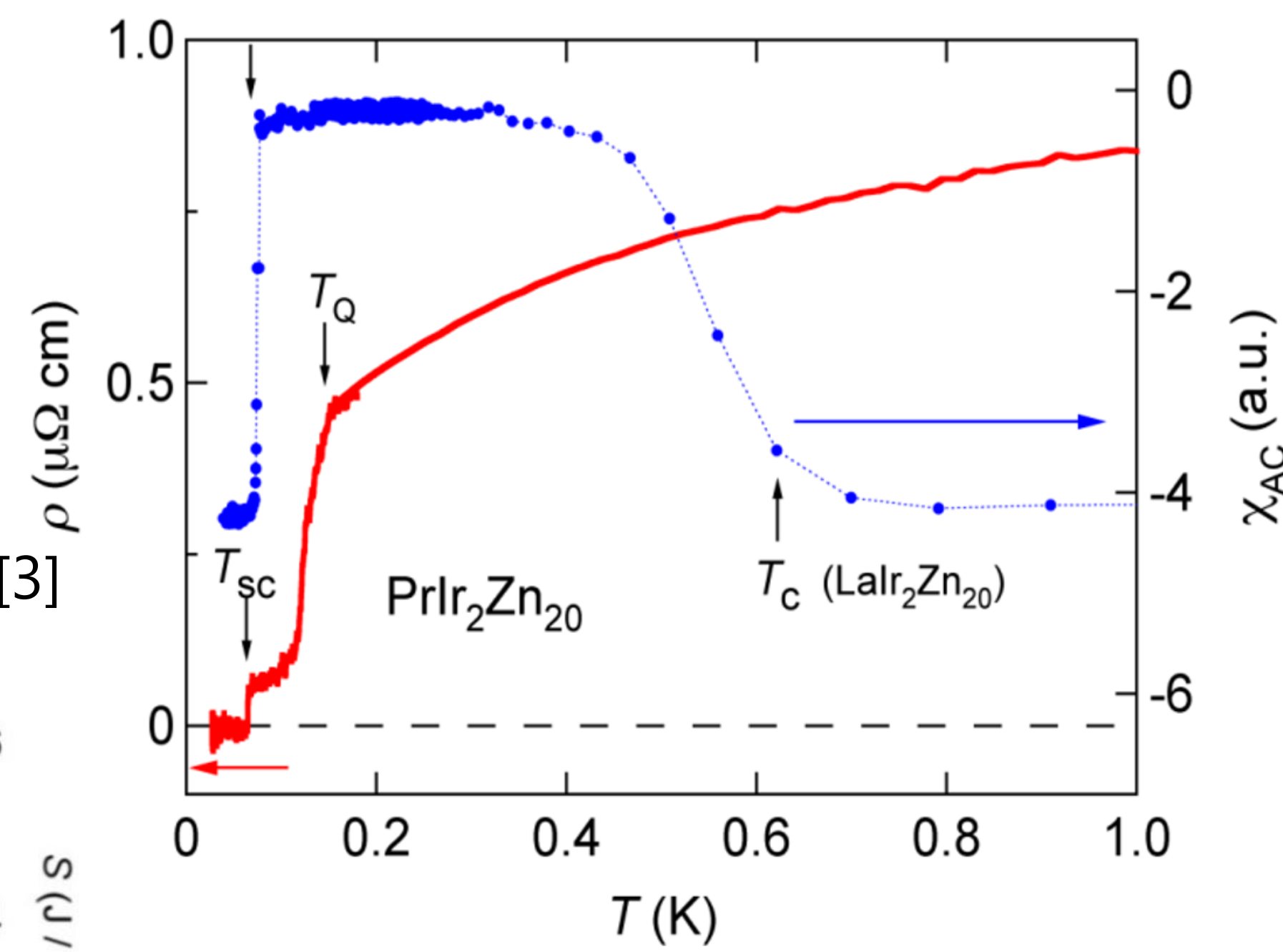
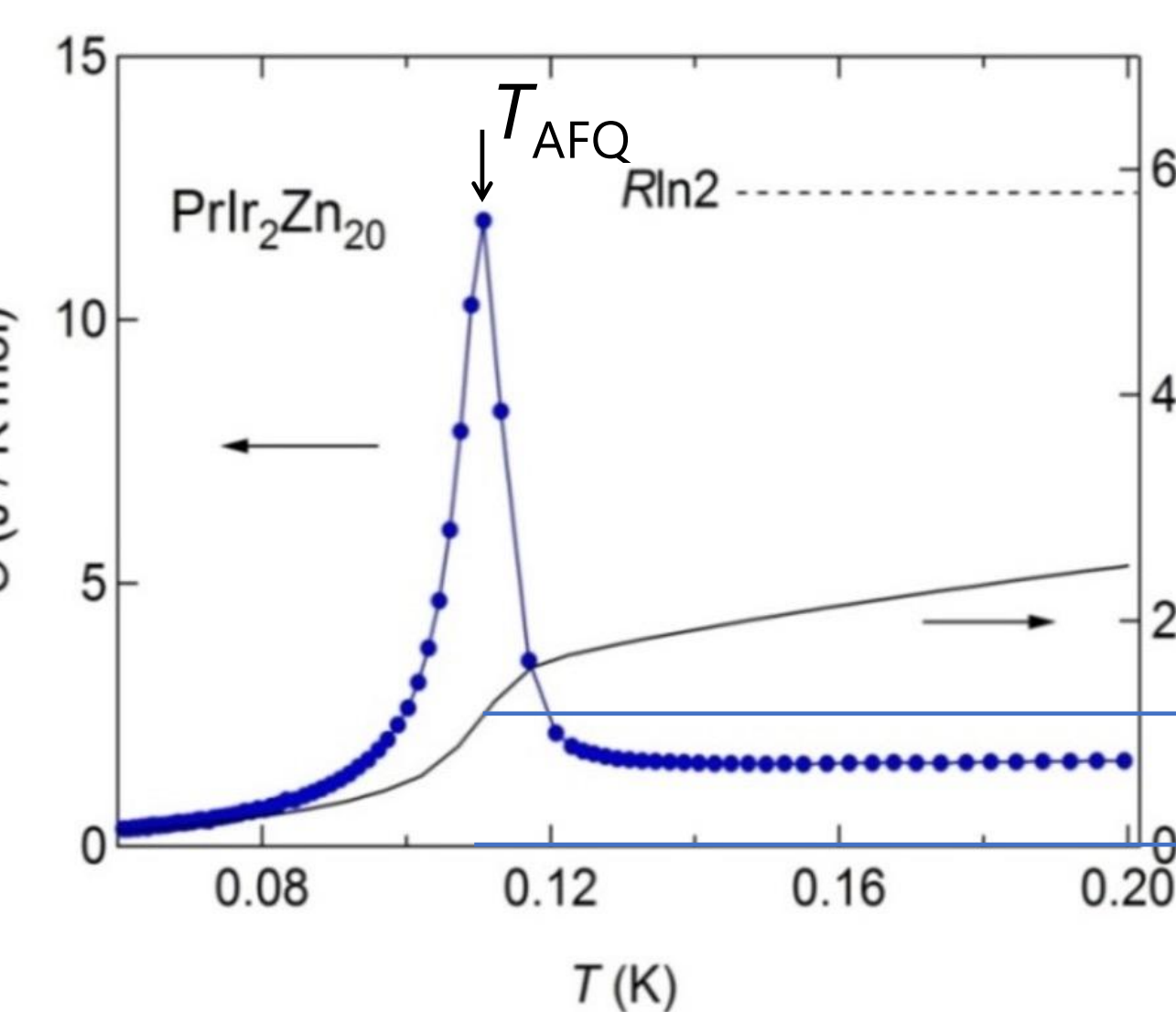


Antiferroquadrupolar (AFQ) Order and Superconducting(SC) Transition

• Temperature dependence of electrical resistivity $\rho(T)$ and AC susceptibility $\chi_{AC}(T)$ [3]

AFQ ordering at $T_{AFQ} = 0.11$ K
SC transition at $T_{SC} = 0.05$ K

• Temperature dependence of specific heat $C(T)$ and entropy $S(T)$ [3]



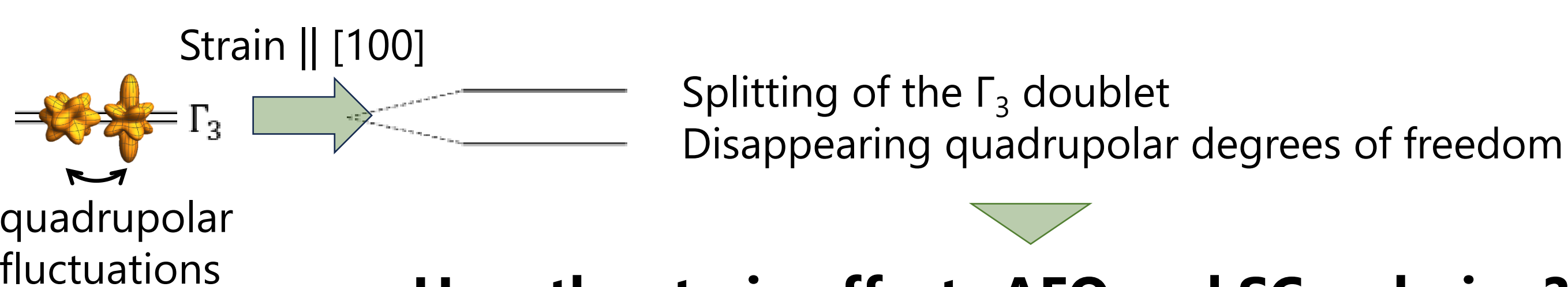
Entropy release at T_{AFQ} reaches only 80% of $R\ln 2$
→ Residual quadrupolar fluctuations below T_{AFQ}

$$\Delta S = 0.2 R \ln 2$$

Do quadrupolar fluctuations contribute to superconducting pairing?

Purpose

Investigate changes in T_{AFQ} and T_{SC} by controlling the direction and magnitude of strain applied to single-crystal $\text{PrIr}_2\text{Zn}_{20}$ to clarify the correlation between AFQ and SC.

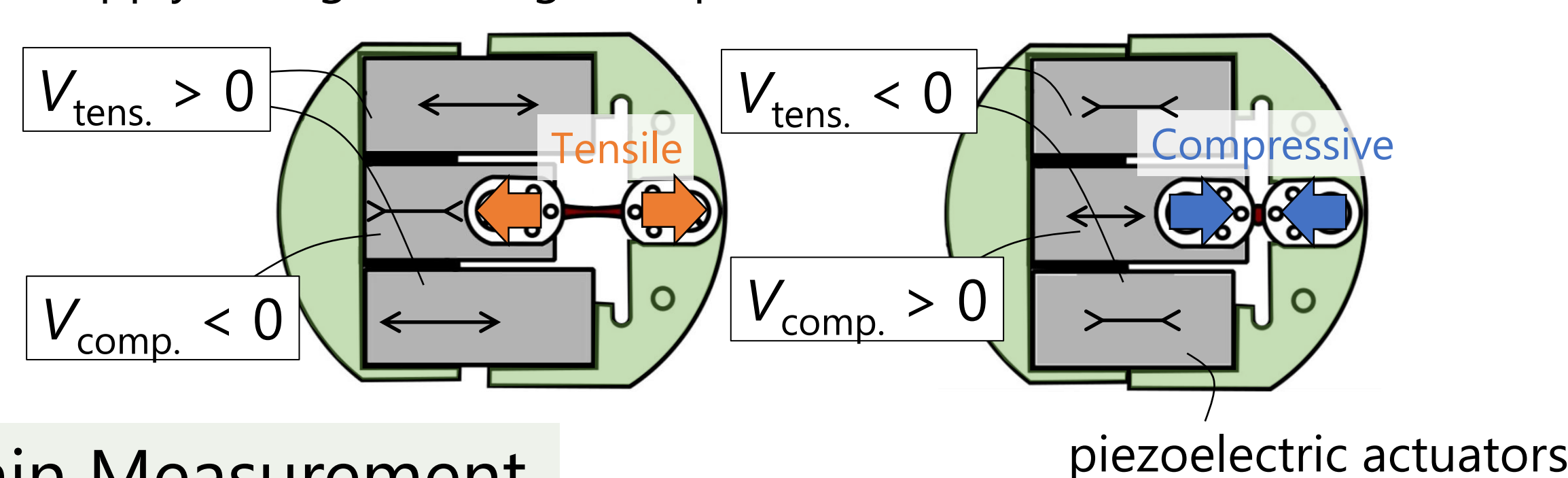


How the strain affects AFQ and SC ordering?

Method

Strain Cell: CS100 (Razorbill Instruments) [4]

Apply voltage to integrated piezoelectric actuators



Strain Measurement

Capacitance variation of the built-in capacitance sensor in the strain cell

$$\varepsilon = \frac{1}{L} C_0 d_0 \Delta \left(\frac{1}{C} \right) \quad C_0 = 1.302 \text{ pF}, d_0 = 43.57 \text{ } \mu\text{m}$$

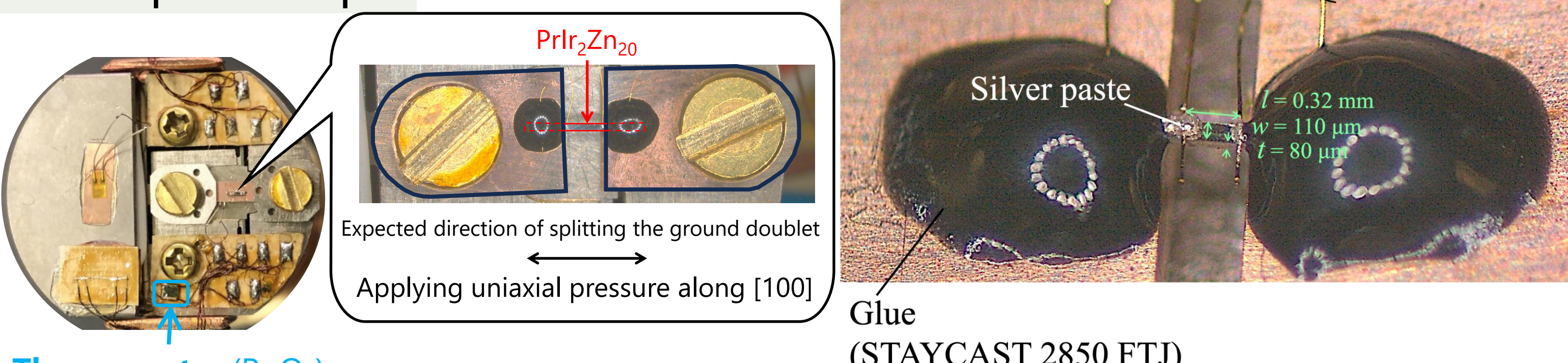
Cooling system

Adiabatic Demagnetization Refrigerator (ADR, Oxford Instruments) $T \geq 0.03$ K

Electrical resistivity measurement

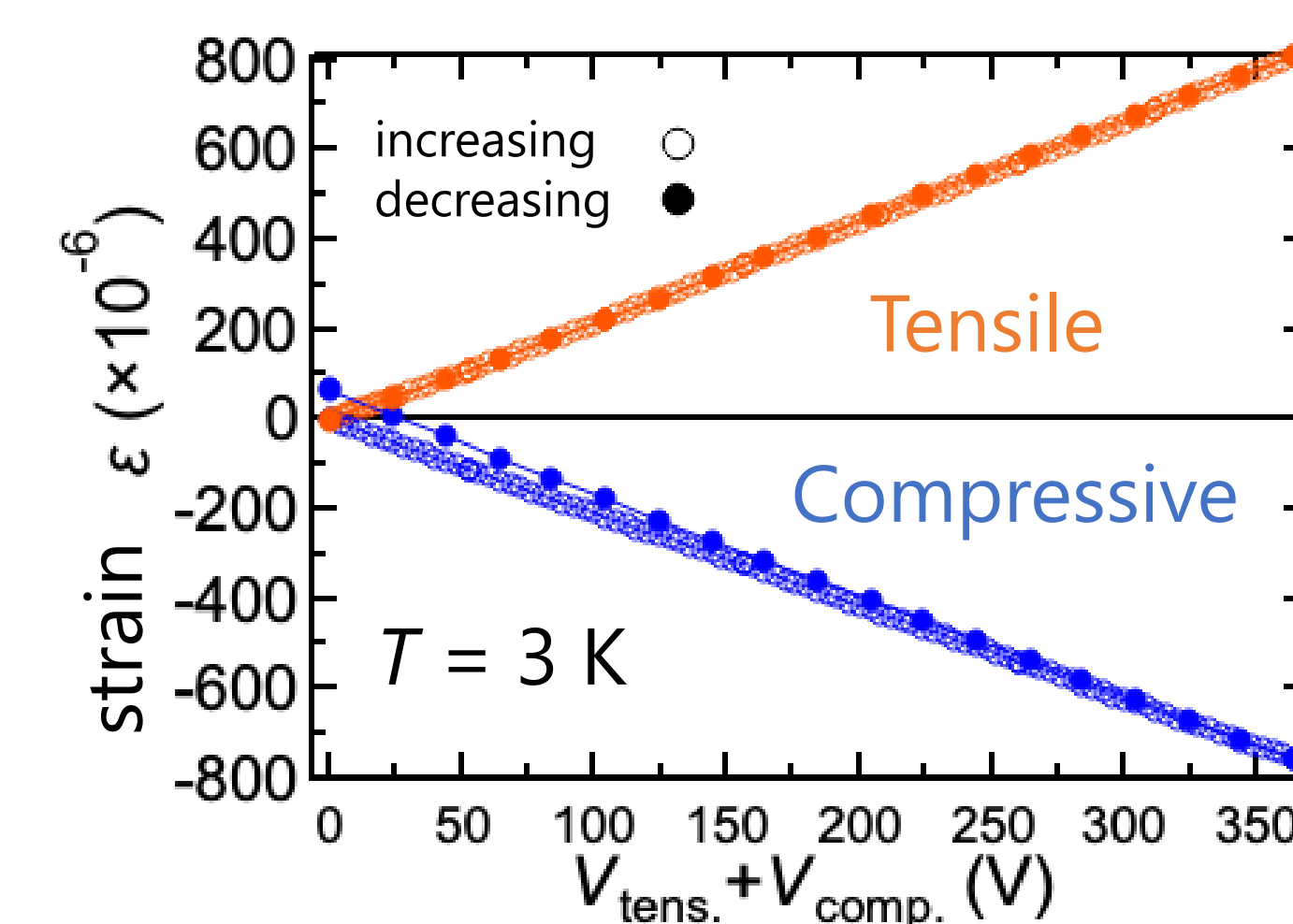
AC 4-terminal method

Sample setup



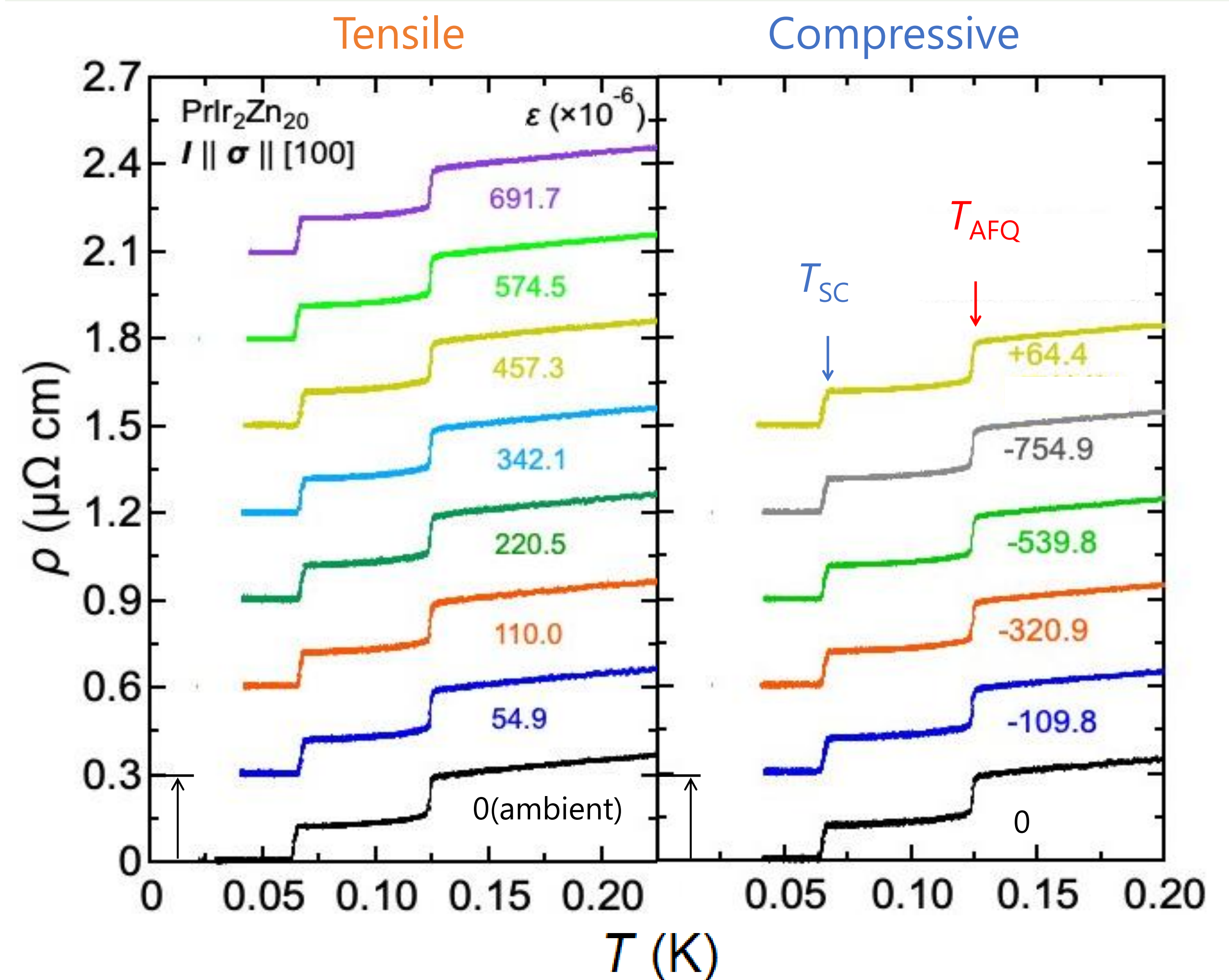
Results

Strain Response to Applied Piezoelectric Voltage



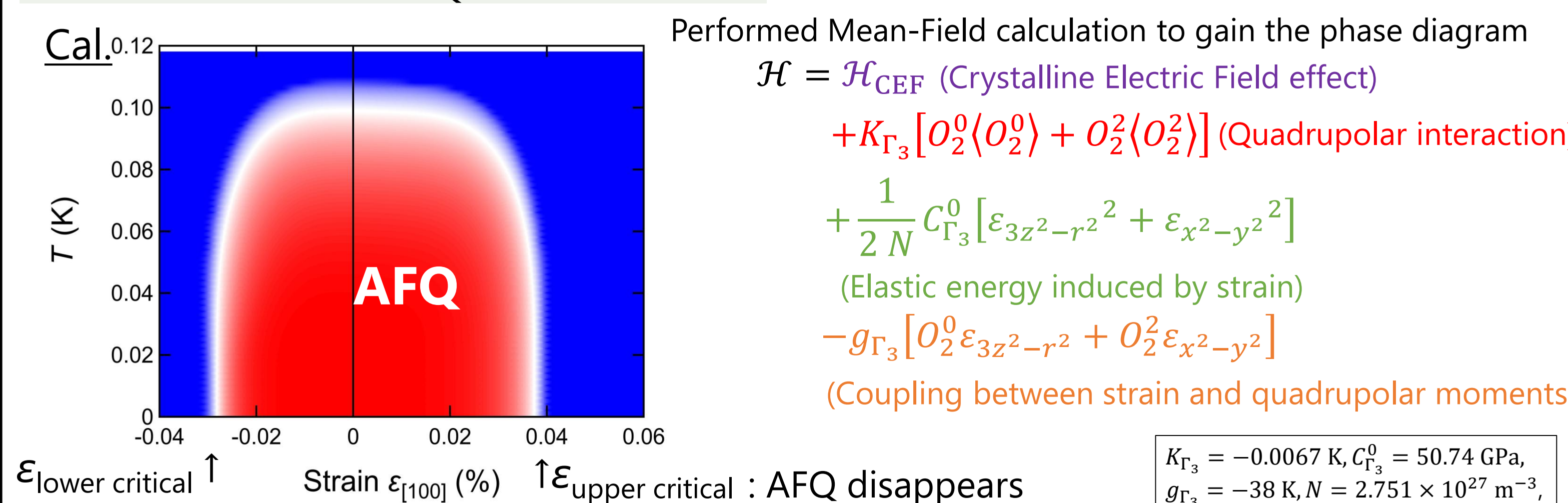
Maximum strain $\varepsilon = 800 \times 10^{-6}$
Hysteresis loop in strain: $\Delta \varepsilon = 60 \times 10^{-6}$

Uniaxial Strain Dependence of Electrical Resistivity in $\text{PrIr}_2\text{Zn}_{20}$



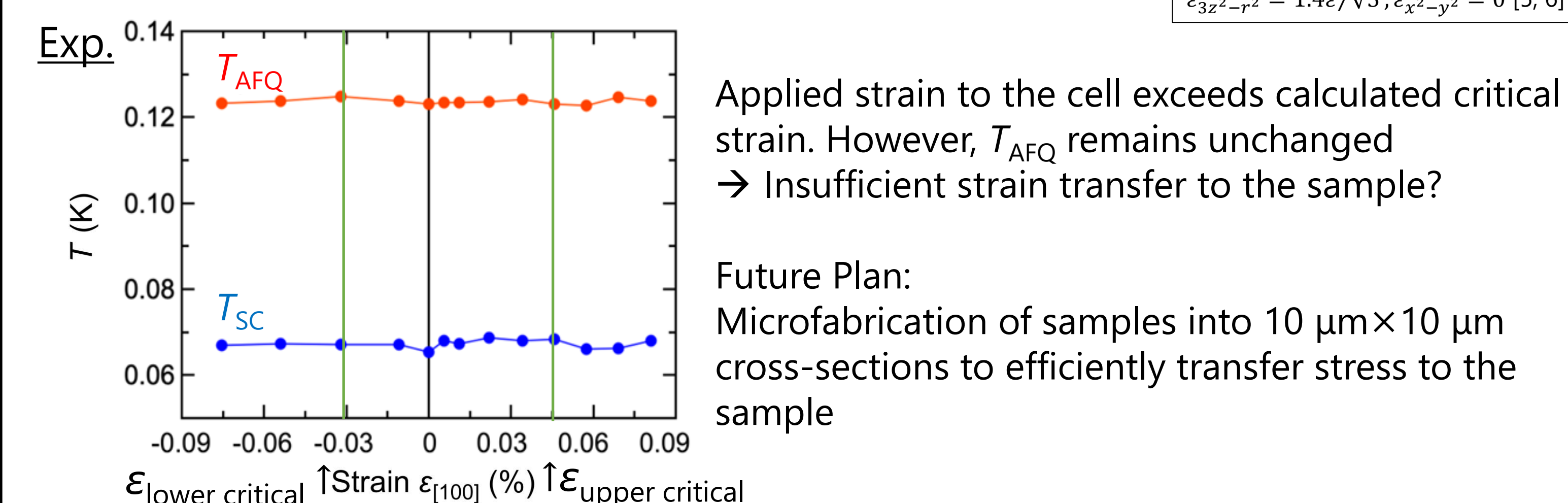
Discussion

Strain vs. T_{AFQ} and T_{SC}



Performed Mean-Field calculation to gain the phase diagram
 $\mathcal{H} = \mathcal{H}_{CEF}$ (Crystalline Electric Field effect)
 $+ K_{\Gamma_3} [O_2^0 \langle O_2^0 \rangle + O_2^2 \langle O_2^2 \rangle]$ (Quadrupolar interaction)
 $+ \frac{1}{2N} C_{\Gamma_3}^0 [\varepsilon_{3z^2-r^2}^2 + \varepsilon_{x^2-y^2}^2]$ (Elastic energy induced by strain)
 $- g_{\Gamma_3} [O_2^0 \varepsilon_{3z^2-r^2} + O_2^2 \varepsilon_{x^2-y^2}]$ (Coupling between strain and quadrupolar moments)

$$K_{\Gamma_3} = -0.0067 \text{ K}, C_{\Gamma_3}^0 = 50.74 \text{ GPa}, g_{\Gamma_3} = -38 \text{ K}, N = 2.751 \times 10^{27} \text{ m}^{-3}, \varepsilon_{3z^2-r^2} = 1.4\varepsilon/\sqrt{3}, \varepsilon_{x^2-y^2} = 0 \text{ [5, 6]}$$



Applied strain to the cell exceeds calculated critical strain. However, T_{AFQ} remains unchanged
→ Insufficient strain transfer to the sample?

Future Plan:
Microfabrication of samples into $10 \text{ } \mu\text{m} \times 10 \text{ } \mu\text{m}$ cross-sections to efficiently transfer stress to the sample

Summary

- Quantitative in-situ control of strain down to $T = 0.03$ K.
- Measured electrical resistivity of $\text{PrIr}_2\text{Zn}_{20}$ under uniaxial strain
- Estimated of critical strain for AFQ Suppression via Mean-Field Theory
- Compared with experimental results suggests insufficient strain transfer to the sample
- Future plan:
 - Focused Ion Beam (FIB) microfabrication at Tohoku University
 - Measurements at Hiroshima University

Reference

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